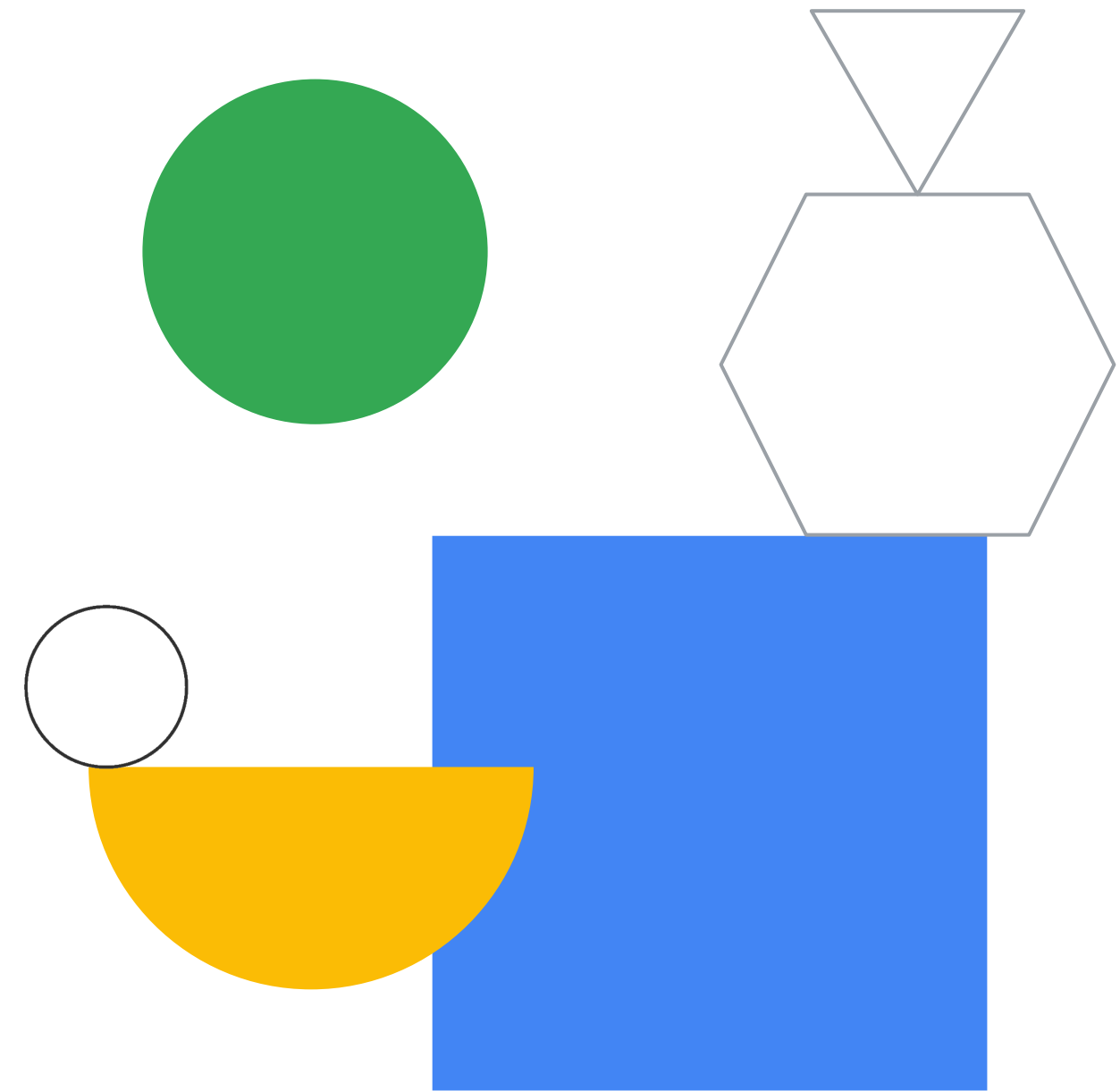


Preparing for Your Associate Cloud Engineer Journey

Module 4: Ensuring Successful Operation of a Cloud Solution



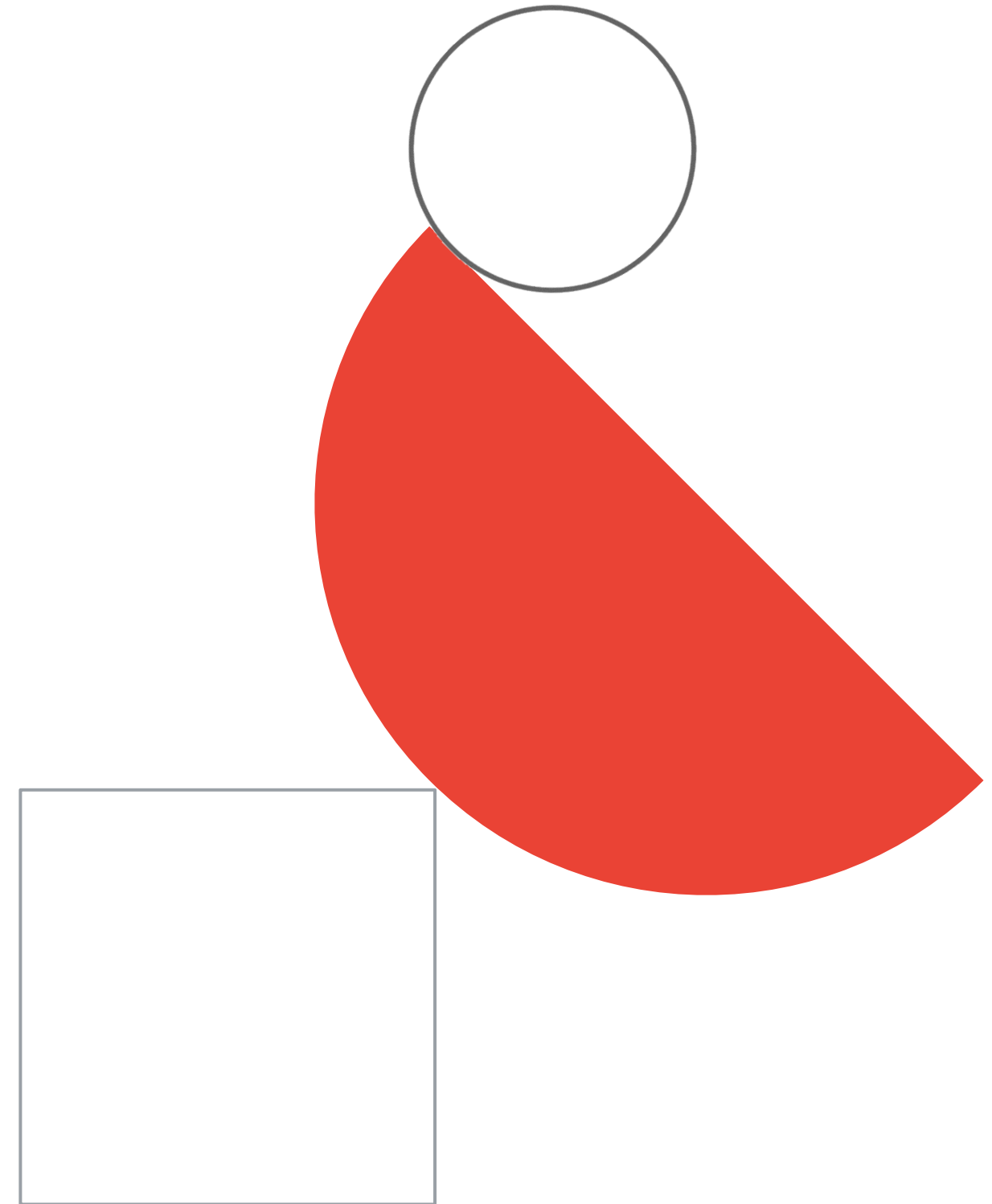


Module agenda

- 01 Managing Cymbal Superstore's cloud solutions
- 02 Diagnostic questions
- 03 Review and study planning

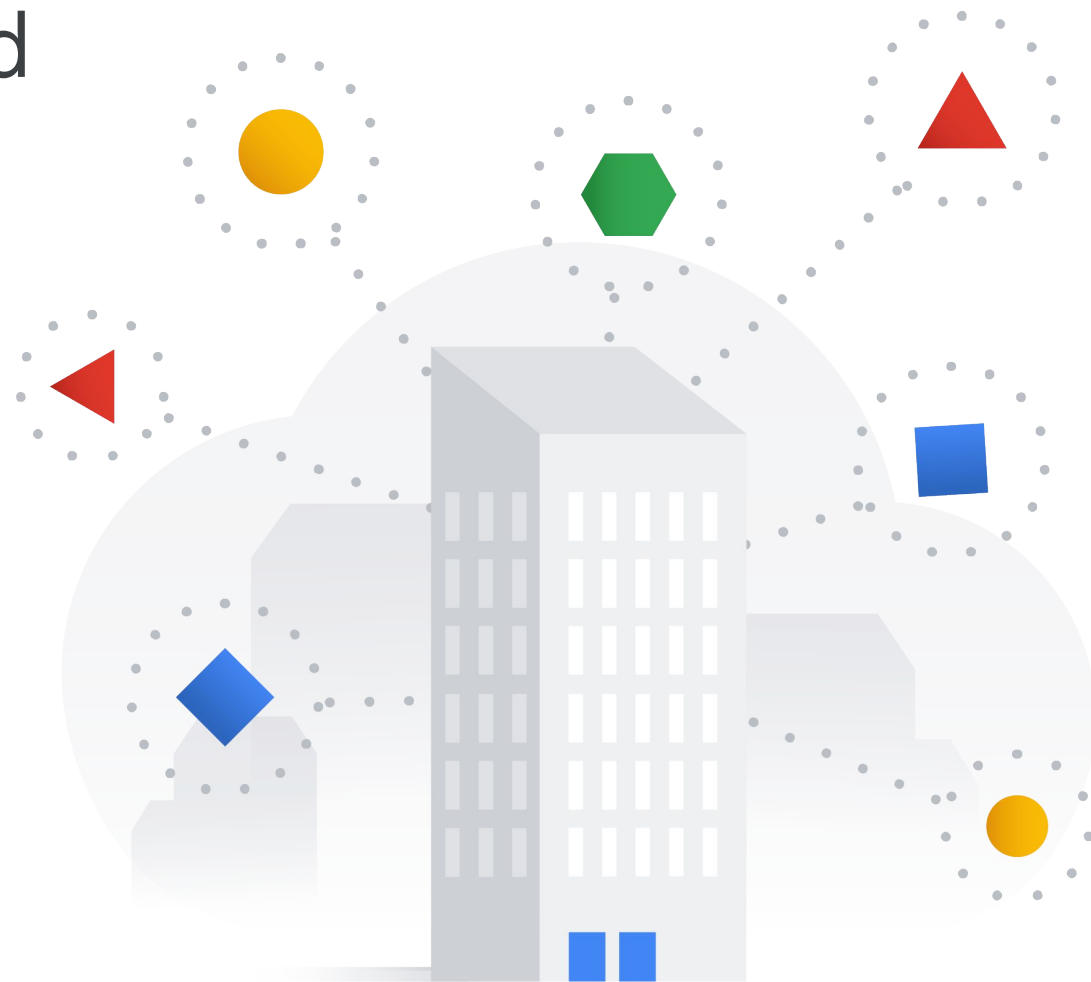


Managing Cymbal Superstore's cloud solutions



The next step:

managing Cymbal
Superstore's cloud
solutions



- Managing Compute Engine resources
- Managing GKE resources
- Managing Cloud Run resources
- Managing storage and database solutions
- Managing networking resources
- Monitoring and logging



Managing Cymbal Superstore's **supply chain app**

Upgrading managed instance groups:

```
gcloud compute instance-groups managed  
rolling-action start_update cymball_supplychain_ig \  
  --version=template=cymball_supplychain_ig_templat  
e_<yymmdd> \  
  --type=proactive\  
  --region=us-central1
```


Managing Cymbal Superstore's **ecommerce app**

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: cymbal-ecommerce-ingress
  annotations:
    # If the class annotation is not specified
    # it defaults to "gce".
    kubernetes.io/ingress.class: "gce"
spec:
  rules:
  - http:
      paths:
      - path: /sales
        pathType: ImplementationSpecific
        backend:
          service:
            name: sales-service
            port:
              number: 60000
```

```
- path: /support
  pathType: ImplementationSpecific
  backend:
    service:
      name: support-service
      port:
        number: 80
```

Here is an example of an ingress object that implements an external layer 7 (http(s)) load balancer

Managing Cymbal Superstore's transportation management app

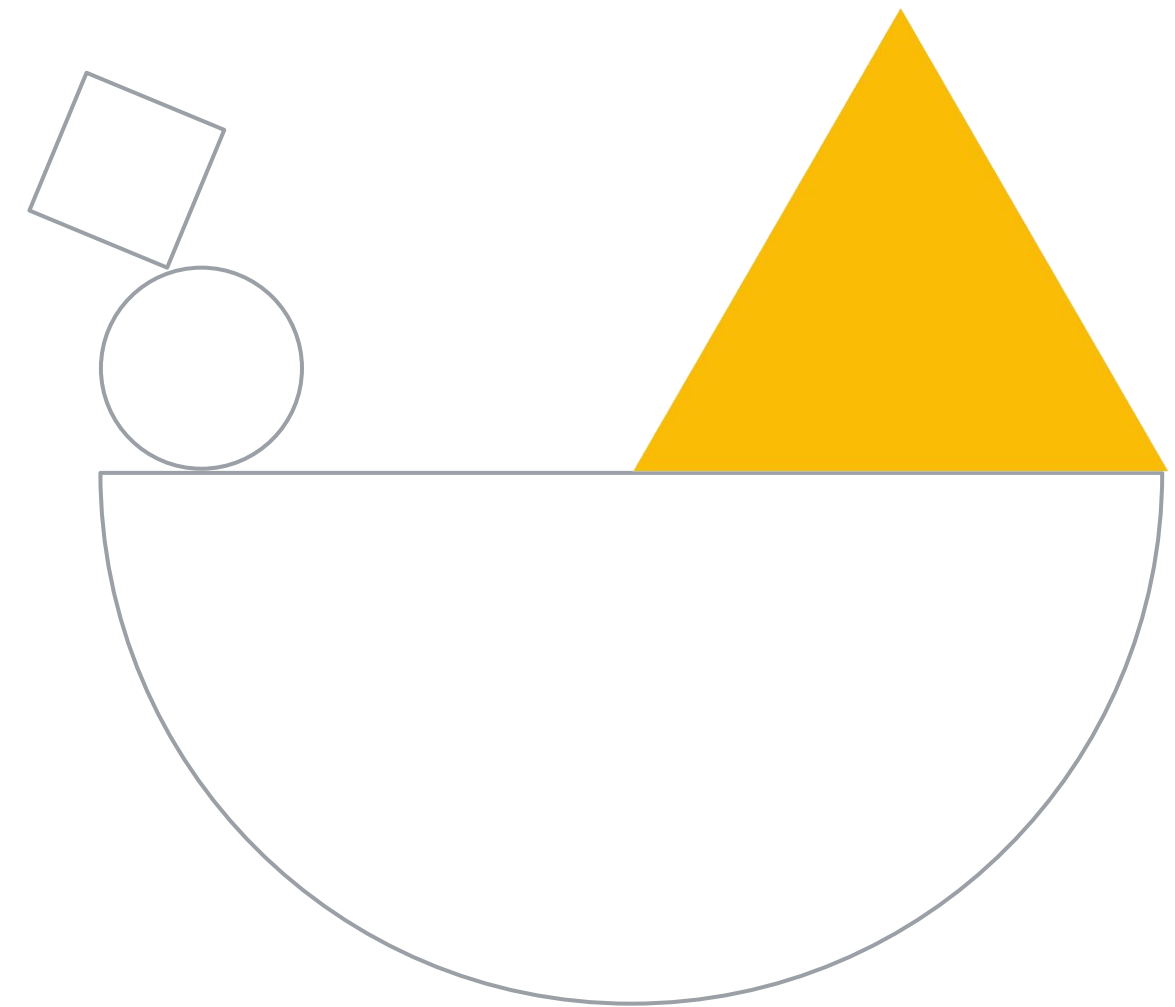
Querying external data such as Bigtable data with BigQuery:

1. create a table definition file
2. create a permanent external table in BigQuery

```
bq mk --external_table_definition=cymbal_trans_mngt_bt_def /  
cymbal_data_set.trans_mngt_ext_tbl
```

3. Query the data using the permanent table reference in the from clause of a sql query

Diagnostic questions

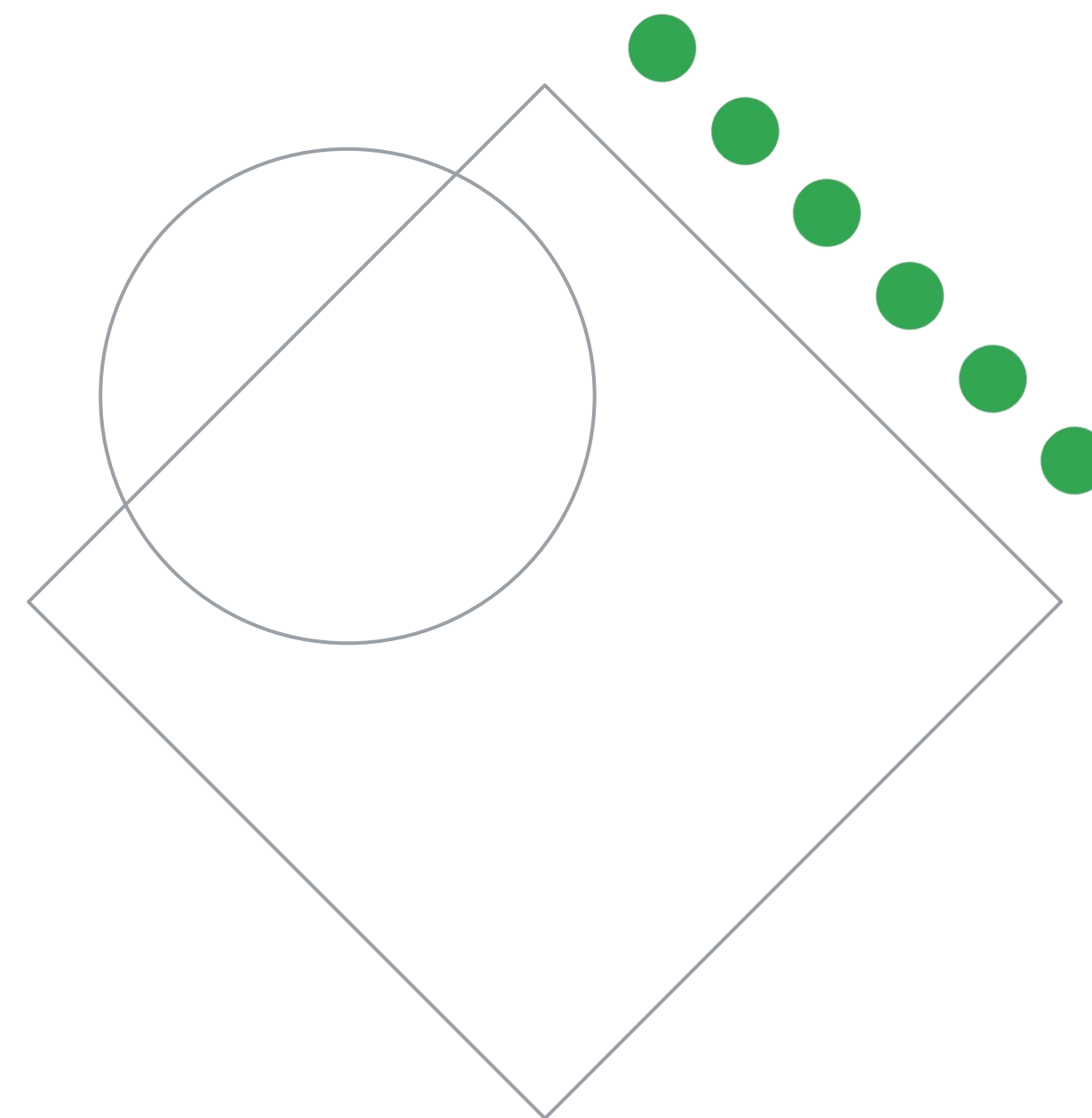


Please complete the diagnostic questions now

- The diagnostic questions are available in the workbook.

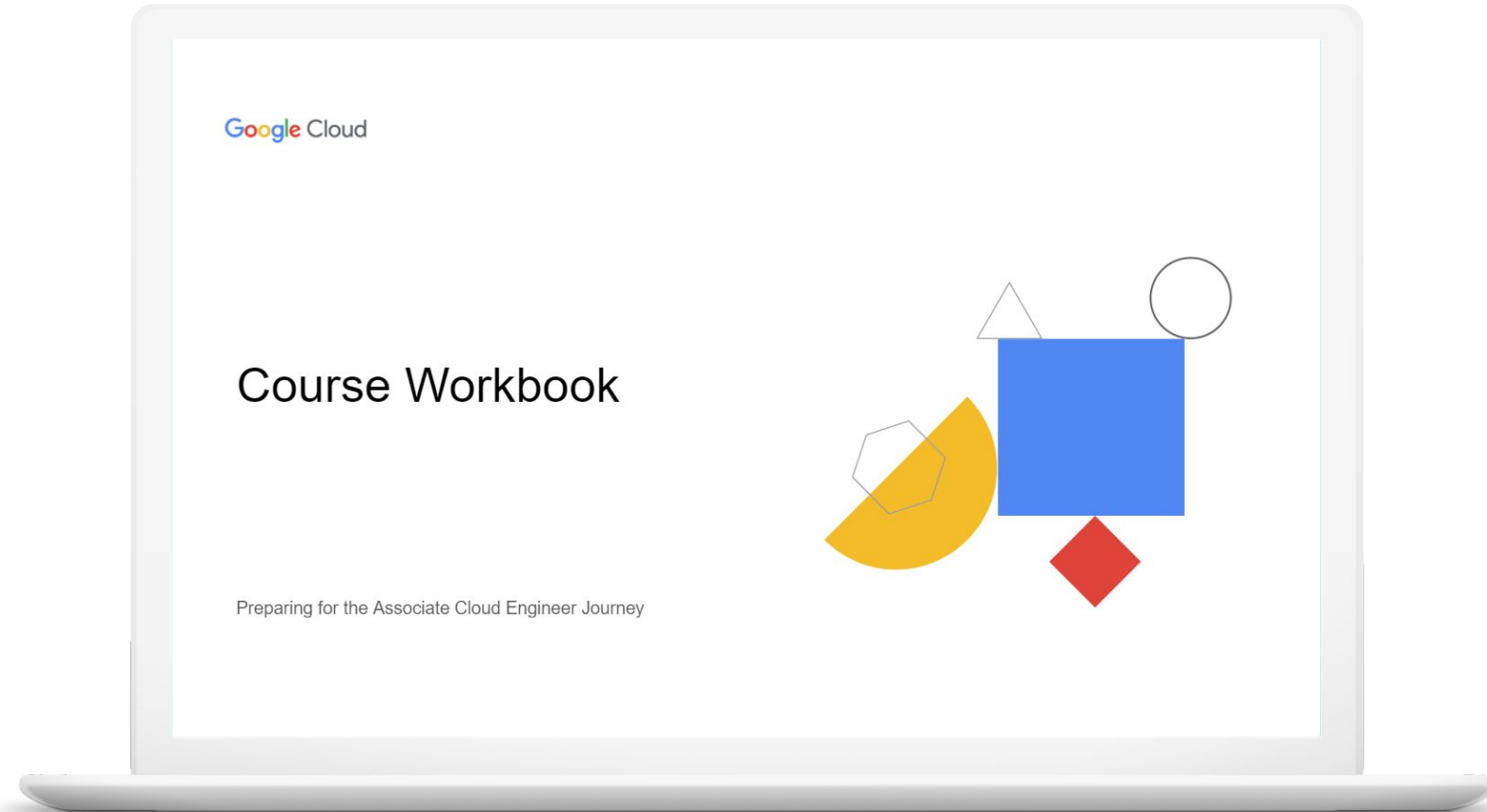


Review and study planning



Your study plan:

Ensuring successful operation of a cloud solution



4.1

Managing Compute Engine resources

4.2

Managing Google Kubernetes Engine resources

4.3

Managing Cloud Run resources

4.4

Managing storage and database solutions

4.5

Managing networking resources

4.6

Monitoring and logging

4.1 | Managing Compute Engine resources

Tasks include:

- Managing a single VM instance (e.g., start, stop, edit configuration, or delete an instance)
- Remotely connecting to the instance
- Attaching a GPU to a new instance and installing necessary dependencies
- Viewing current running VM inventory (instance IDs, details)
- Working with snapshots (e.g., create a snapshot from a VM, view snapshots, delete a snapshot)
- Working with images (e.g., create an image from a VM or a snapshot, view images, delete an image)
- Working with instance groups (e.g., set autoscaling parameters, assign instance template, create an instance template, remove instance group)
- Working with management interfaces (e.g., Cloud Console, Cloud Shell, Cloud SDK)

4.1 | Diagnostic Question 01 Discussion



You want to view a description of your available snapshots using the command line interface (CLI). What gcloud command should you use?

- A. gcloud compute snapshots list
- B. gcloud snapshots list
- C. gcloud compute snapshots get
- D. gcloud compute list snapshots

4.1 | Diagnostic Question 01 Discussion



You want to view a description of your available snapshots using the command line interface (CLI). What gcloud command should you use?

- A. **gcloud compute snapshots list**
- B. gcloud snapshots list
- C. gcloud compute snapshots get
- D. gcloud compute list snapshots



Getting information about snapshots

To list Compute Engine disk snapshots:

```
gcloud compute snapshots list --project PROJECT_ID
```

To describe snapshots:

```
gcloud compute snapshots describe SNAPSHOT_NAME
```

4.1 | Diagnostic Question 02 Discussion



You have a scheduled snapshot you are trying to delete, but the operation returns an error.

What should you do to resolve this problem?

- A. Delete the downstream incremental snapshots before deleting the main reference.
- B. Delete the object the snapshot was created from.
- C. Detach the snapshot schedule before deleting it.
- D. Restore the snapshot to a persistent disk before deleting it.

4.1 | Diagnostic Question 02 Discussion



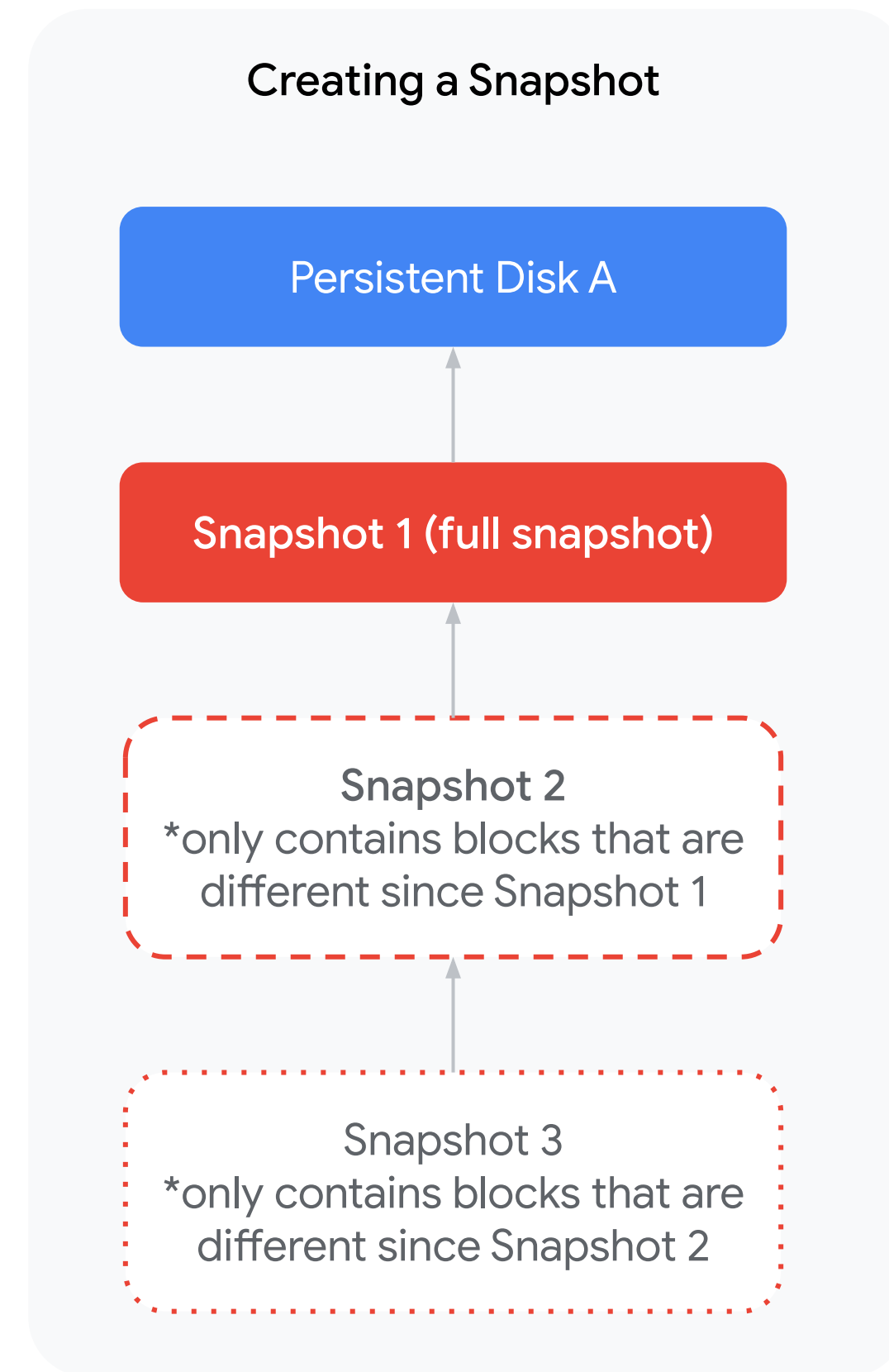
You have a scheduled snapshot you are trying to delete, but the operation returns an error.

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- A. Delete the downstream incremental snapshots before deleting the main reference.
- B. Delete the object the snapshot was created from.
- C. Detach the snapshot schedule before deleting it.**
- D. Restore the snapshot to a persistent disk before deleting it.



Snapshots are
incremental



4.1 | Diagnostic Question 03 Discussion



Which of the following tasks are part of the process when configuring a managed instance group? (Pick two.)

- A. Defining Health checks.
- B. Providing Number of instances.
- C. Specifying Persistent disks.
- D. Choosing instance Machine type.
- E. Configuring the operating system.

4.1 | Diagnostic Question 03 Discussion



Which of the following tasks are part of the process when configuring a managed instance group? (Pick two.)

- A. Defining Health checks.
- B. Providing Number of instances.
- C. Specifying Persistent disks.
- D. Choosing instance Machine type.
- E. Configuring the operating system.



Implementing an instance group

01

Step 01

Create instance template
e.g. identify machine
type and boot disk

02

Step 02

Configure your instance
group e.g. number of
instances and
autoscaling settings

4.1

Managing Compute Engine resources

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M3 Virtual Machines and Networks in the Cloud

[Architecting with Google Compute Engine](#)

- M3 Virtual Machines
- M9 Load Balancing and Autoscaling



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[Essential Google Cloud Infrastructure: Foundation](#)

- M3 Virtual Machines

[Elastic Google Cloud Infrastructure: Scaling and Automation](#)

- M2 Load Balancing and Autoscaling



Documentation

[Working with persistent disk snapshots | Compute Engine Documentation](#)

[Working with persistent disk snapshots | Compute Engine Documentation](#)

[Persistent disk snapshots | Compute Engine Documentation](#)

[Instance templates | Compute Engine Documentation](#)

[Instance groups | Compute Engine Documentation](#)

4.2 | Managing Google Kubernetes Engine resources

Tasks include:

- Viewing current running cluster inventory (nodes, pods, services)
- Browsing Docker images and viewing their details in the Artifact Registry
- Working with node pools (e.g., add, edit, or remove a node pool)
- Working with pods (e.g., add, edit, or remove pods)
- Working with services (e.g., add, edit, or remove a service)
- Working with stateful applications (e.g. persistent volumes, stateful sets)
- Managing Horizontal and Vertical autoscaling configurations.
- Working with management interfaces (e.g., Cloud Console, Cloud Shell, Cloud SDK, kubectl)

4.2 | Diagnostic Question 04 Discussion



Cymbal Superstore's GKE cluster requires an internal http(s) load balancer. You are creating the configuration files required for this resource.

What is the proper setting for this scenario?

- A. Annotate your ingress object with an `ingress.class` of "gce."
- B. Configure your service object with a type: `LoadBalancer`.
- C. Annotate your service object with a "neg" reference.
- D. Implement custom static routes in your VPC.

4.2 | Diagnostic Question 04 Discussion



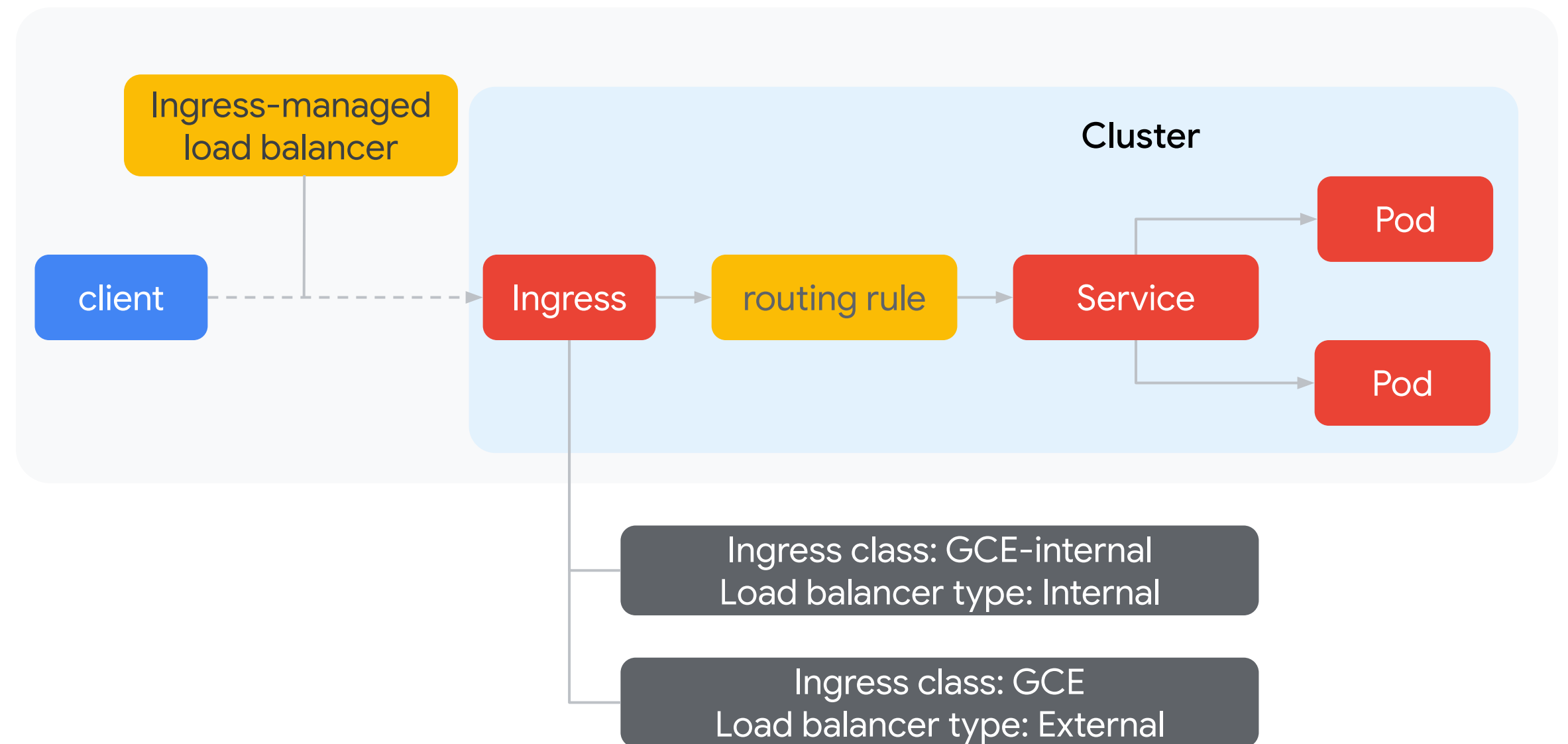
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- B. Configure your service object with a type: `LoadBalancer`.
- C. Annotate your service object with a "neg" reference.
- D. Implement custom static routes in your VPC.



Internal vs External load balancing in Kubernetes



4.2 | Diagnostic Question 05 Discussion



What Kubernetes object provides access to logic running in your cluster via endpoints that you define?

- A. Pod templates
- B. Pods
- C. Services
- D. Deployments

4.2 | Diagnostic Question 05 Discussion

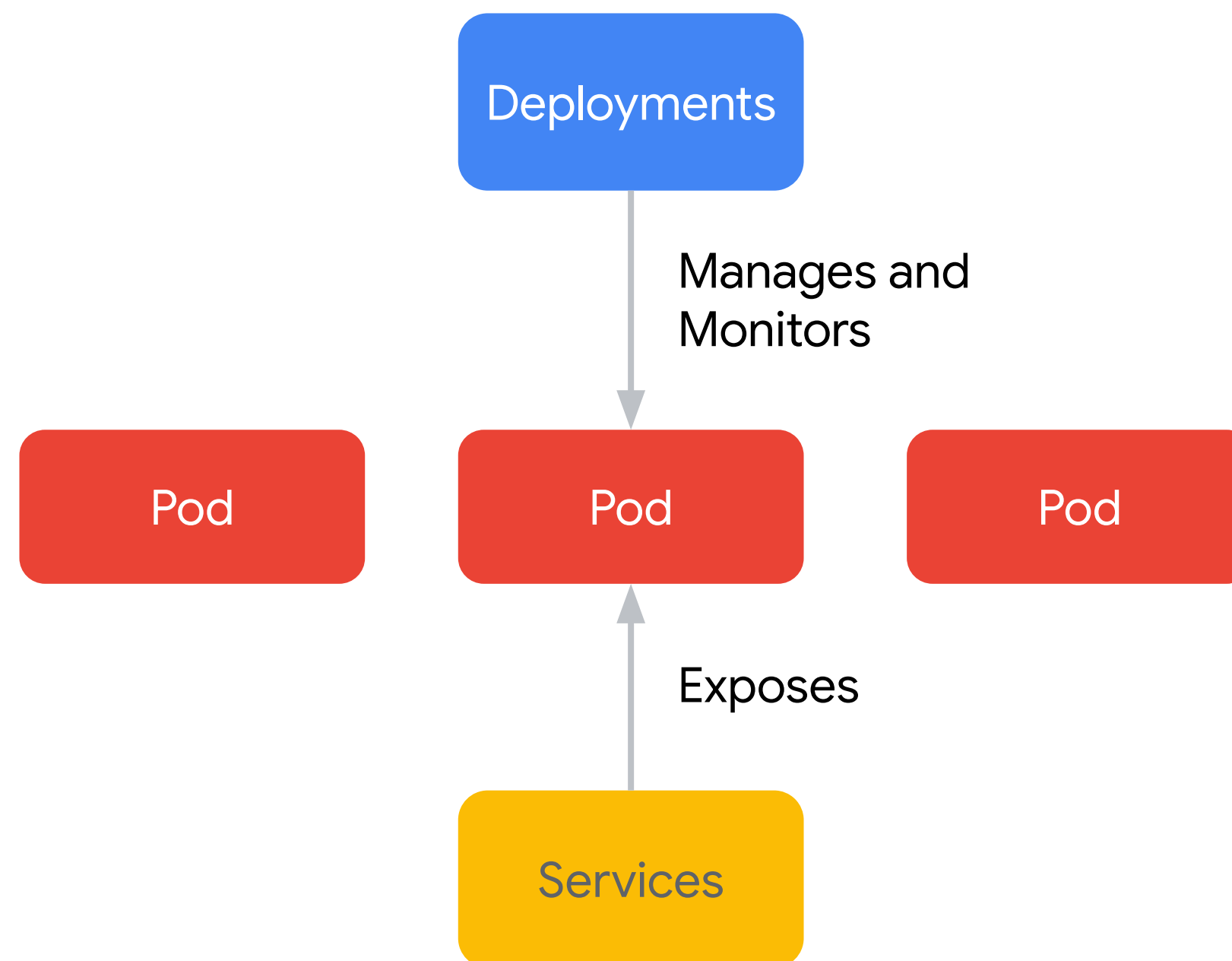


What Kubernetes object provides access to logic running in your cluster via endpoints that you define?

- A. Pod templates
- B. Pods
- C. Services**
- D. Deployments



Kubernetes objects



4.2 | Diagnostic Question 06 Discussion

What is the declarative way to initialize and update Kubernetes objects?

- A. kubectl apply
- B. kubectl create
- C. kubectl replace
- D. kubectl run



4.2 | Diagnostic Question 06 Discussion

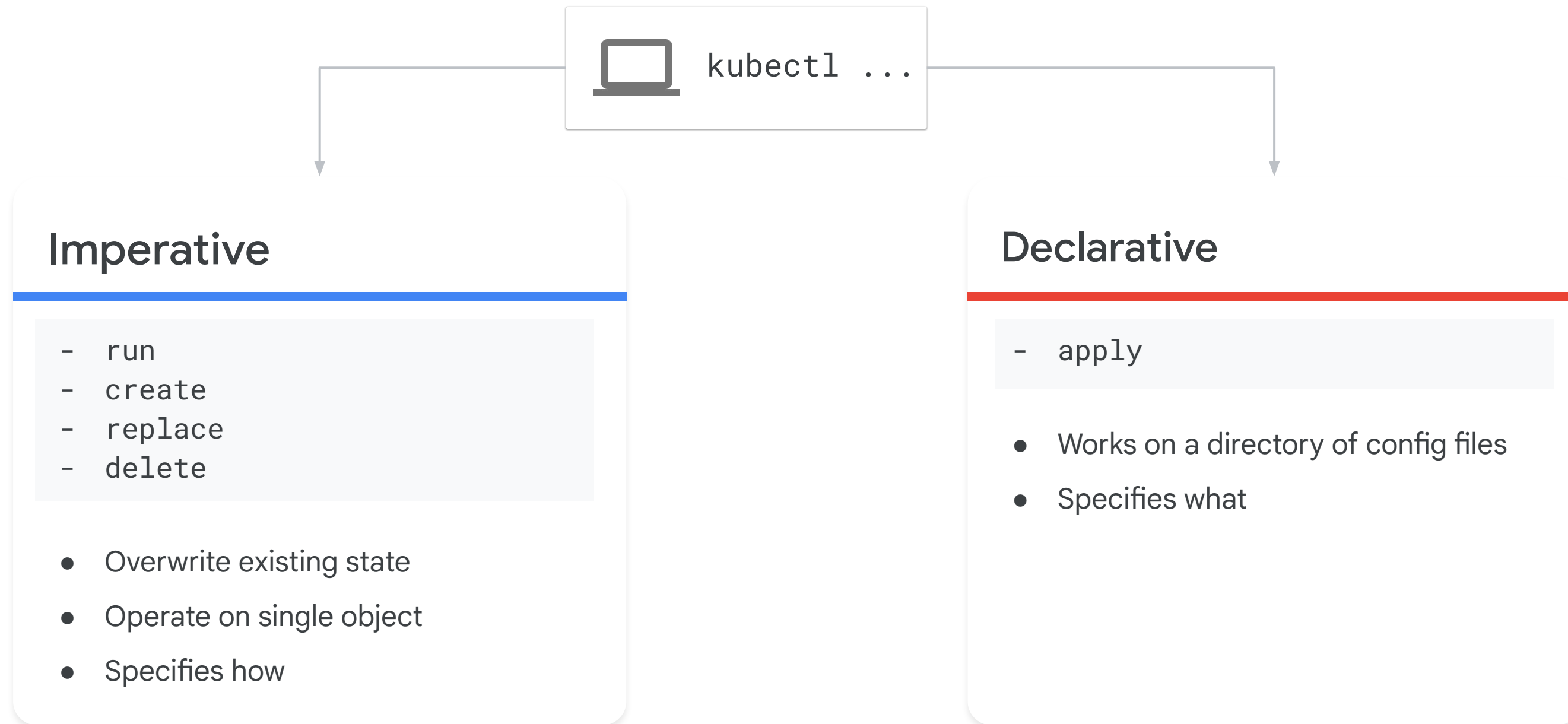


What is the declarative way to initialize and update Kubernetes objects?

- A. `kubectl apply`
- B. `kubectl create`
- C. `kubectl replace`
- D. `kubectl run`



Types of kubectl commands



4.2

Managing Google Kubernetes Engine resources

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M5 Containers in the Cloud

[Getting Started with Google Kubernetes Engine](#)

- M3 Kubernetes Architecture
- M4 Kubernetes Operations

Skill Badge



Google Cloud

[Develop your Google Cloud Network](#)

Documentation

[Ingress for Internal HTTP\(S\) Load Balancing](#)

[Ingress for External HTTP\(S\) Load Balancing](#)

[Configuring Ingress for external load balancing](#)

[Configuring Ingress for Internal HTTP\(S\) Load Balancing](#)

[GKE overview | Kubernetes Engine Documentation](#)

[Pod | Kubernetes Engine Documentation](#)

[Deployment | Kubernetes Engine Documentation](#)

[Services | Kubernetes Engine Documentation](#)

[Overview of deploying workloads | Kubernetes Engine Documentation](#)

[Kubernetes Object Management](#)

4.3 | Managing Cloud Run resources

Tasks include:

- Adjusting application traffic splitting parameters
- Setting scaling parameters for autoscaling instances
- Determining whether to run Cloud Run (fully managed) or Cloud Run for Anthos

4.3 | Diagnostic Question 07 Discussion



You have a Cloud Run service with a database backend. You want to limit the number of connections to your database.

- A. Set Min instances.
- B. Set Max instances.
- C. Set CPU Utilization.
- D. Set Concurrency settings.

What should you do?

4.3 | Diagnostic Question 07 Discussion



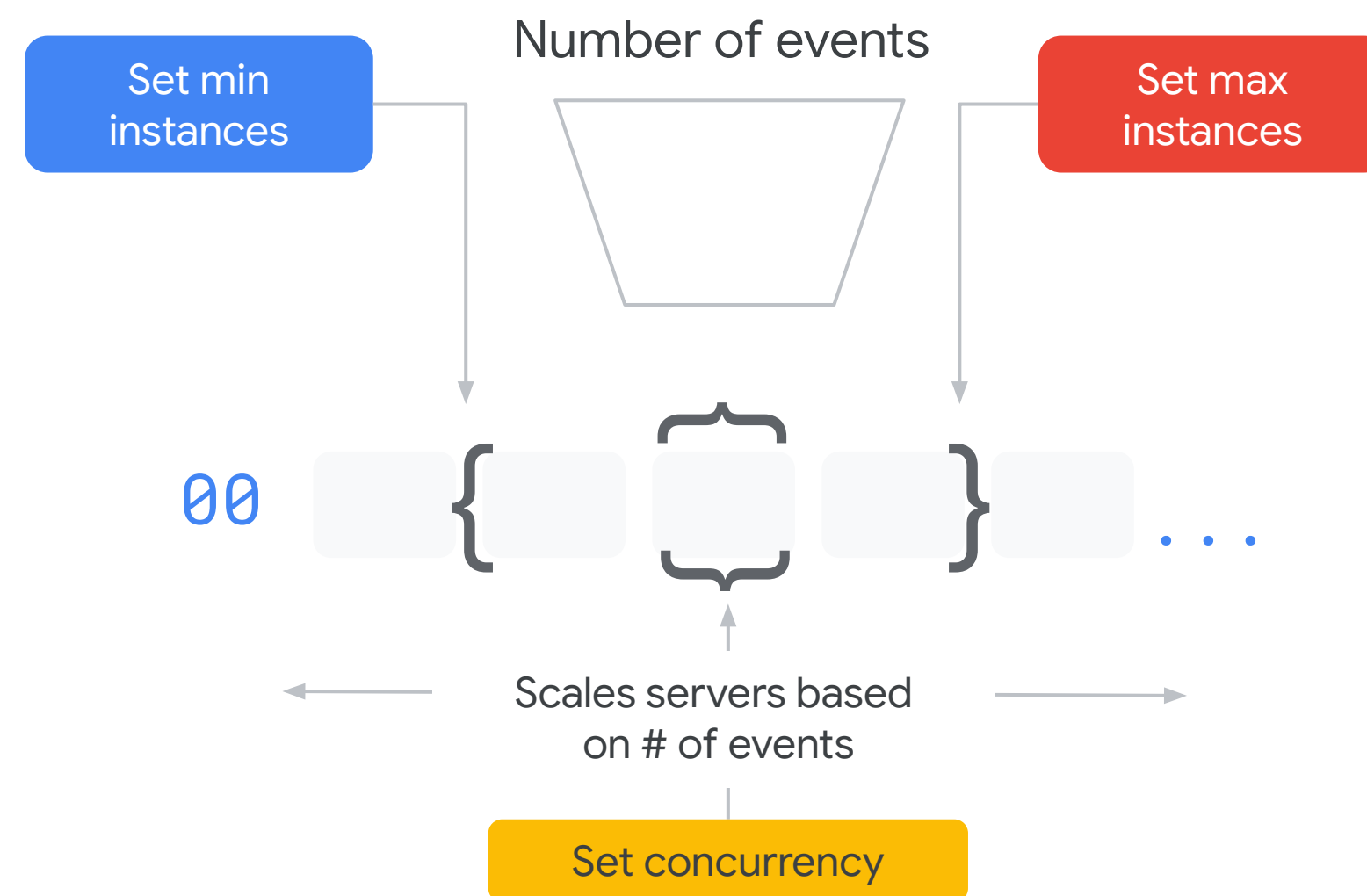
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What should you do?

- A. Set Min instances.
- B. Set Max instances.**
- C. Set CPU Utilization.
- D. Set Concurrency settings.



Cloud Run autoscaling



4.3 | Managing Cloud Run resources

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M6 Applications in the Cloud

Documentation

[About container instance autoscaling | Cloud Run Documentation](#)

4.4 | Managing storage and database solutions

Tasks include:

- Managing and securing objects in and between Cloud Storage buckets
- Setting object life cycle management policies for Cloud Storage buckets
- Executing queries to retrieve data from data instances (e.g., Cloud SQL, BigQuery, Spanner, Datastore, Bigtable)
- Estimating costs of data storage resources
- Backing up and restoring database instances (e.g., Cloud SQL, Datastore)
- Reviewing job status in Dataproc, Dataflow, or BigQuery

4.4 | Diagnostic Question 08 Discussion



You want to implement a lifecycle rule that changes your storage type from Standard to Nearline after a specific date.

What conditions should you use?
(Pick two.)

- A. Age
- B. CreatedBefore
- C. MatchesStorageClass
- D. IsLive
- E. NumberofNewerVersions

4.4 | Diagnostic Question 08 Discussion



You want to implement a lifecycle rule that changes your storage type from Standard to Nearline after a specific date.

What conditions should you use?
(Pick two.)

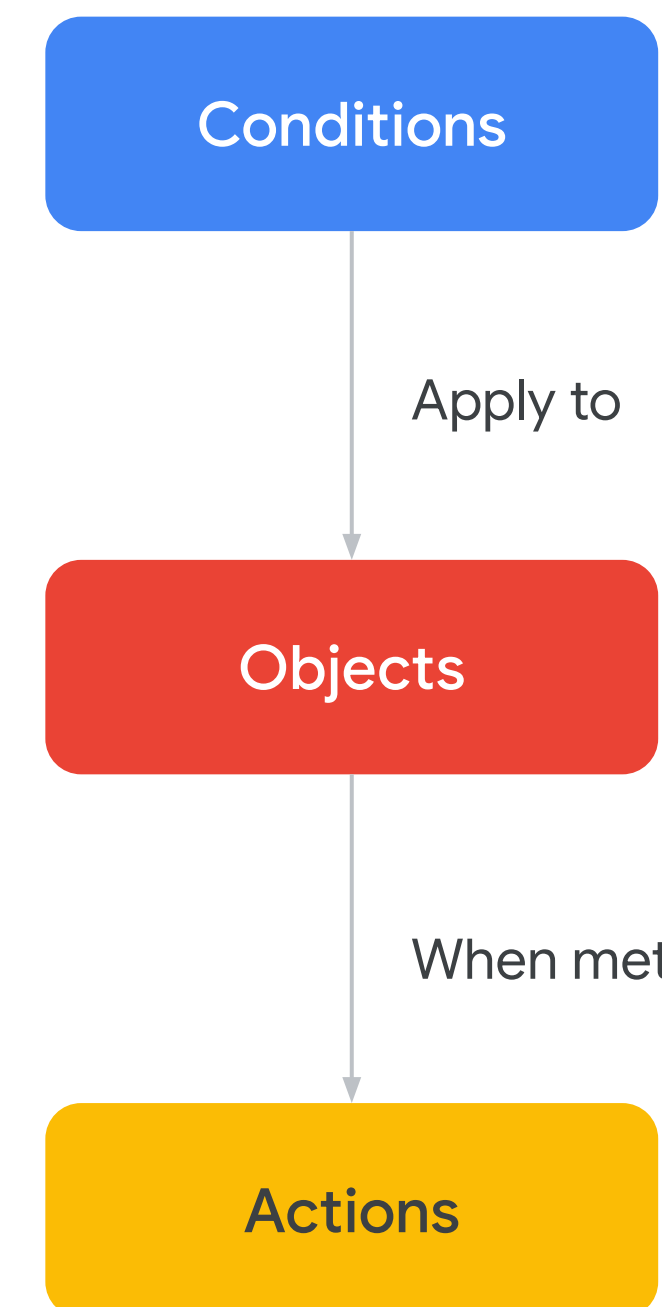
- A. Age
- B. CreatedBefore**
- C. MatchesStorageClass**
- D. IsLive
- E. NumberofNewerVersions



Cloud Storage Lifecycle Actions

- Age
- Createdbefore
- Customtimebefore
- ...

- Delete
- SetStorageClass
- ...



4.4 | Managing storage and database solutions

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M4 Storage in the Cloud

[Architecting with Google Compute Engine](#)

- M5 Storage and Database Services



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[Essential Google Cloud Infrastructure: Core Services](#)

- M2 Storage and Database Services



Documentation

[Object Lifecycle Management | Cloud Storage](#)

4.5 | Managing networking resources

Tasks include:

- Adding a subnet to an existing VPC
- Expanding a subnet to have more IP addresses
- Reserving static external or internal IP addresses
- Working with CloudDNS, CloudNAT, Load Balancers and firewall rules

4.5 | Diagnostic Question 09 Discussion



Cymbal Superstore has a subnetwork called mysubnet with an IP range of 10.1.2.0/24. You need to expand this subnet to include enough IP addresses for at most 2000 users or devices.

What should you do?

- A. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 20`
- B. `gcloud networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`
- C. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`
- D. `gcloud compute networks subnets expand-ip-range mysubnet --region us-cetnral1 --prefix-length 22`

4.5 | Diagnostic Question 09 Discussion



Cymbal Superstore has a subnetwork called mysubnet with an IP range of 10.1.2.0/24. You need to expand this subnet to include enough IP addresses for at most 2000 users or devices.

What should you do?

- A. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 20`
- B. `gcloud networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`
- C. **`gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`**
- D. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 22`



Expand IP addresses in a subnet

Current IP address range

Reduce your mask:
e.g. 24 to 20

Expanded address range

4.5 | Managing networking resources

Courses

[Architecting with Google Compute Engine](#)

- M2 Virtual Networks



=

[Essential Google Cloud Infrastructure: Foundation](#)

- M2 Virtual Networks



Documentation

[gcloud compute networks subnets expand-ip-range](#)
[Using VPC networks](#)

4.6 | Monitoring and logging

Tasks include:

- Creating Cloud Monitoring alerts based on resource metrics
- Creating and ingesting Cloud Monitoring custom metrics (e.g., from applications or logs)
- Configuring log sinks to export logs to external systems (e.g., on-premises or BigQuery)
- Configuring logs routers
- Viewing and filtering logs in Cloud Logging
- Viewing specific log message details in Cloud Logging
- Using cloud diagnostics to research an application issue (e.g., viewing Cloud Trace data)
- Viewing Google Cloud status

4.6 | Diagnostic Question 10 Discussion

Cymbal Superstore's supply chain management system has been deployed and is working well. You are tasked with monitoring the system's resources so you can react quickly to any problems. You want to ensure the CPU usage of each of your Compute Engine instances in us-central1 remains below 60%. You want an incident created if it exceeds this value for 5 minutes. You need to configure the proper alerting policy for this scenario.

What should you do?

- A. Choose resource type of VM instance and metric of CPU load, condition trigger if any time series violates, condition is below, threshold is .60, for 5 minutes.
- B. Choose resource type of VM instance and metric of CPU utilization, condition trigger all time series violates, condition is above, threshold is .60 for 5 minutes.
- C. Choose resource type of VM instance, and metric of CPU utilization, condition trigger if any time series violates, condition is below, threshold is .60 for 5 minutes.
- D. Choose resource type of VM instance and metric of CPU utilization, condition trigger if any time series violates, condition is above, threshold is .60 for 5 minutes.



4.6 | Diagnostic Question 10 Discussion

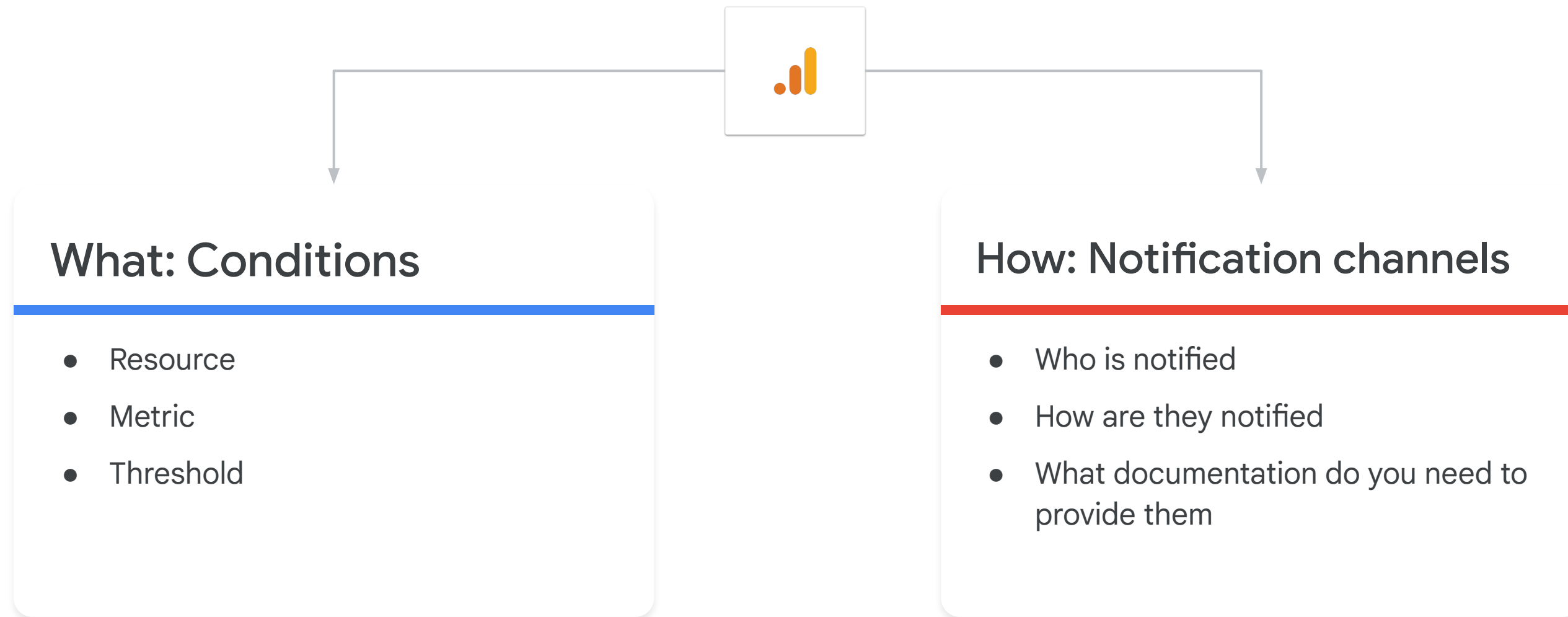
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- D. Choose resource type of VM instance and metric of CPU utilization, condition trigger if any time series violates, condition is above, threshold is .60 for 5 minutes.**



Cloud operations custom alerts



4.6 | Monitoring and logging

Courses

[Architecting with Google Compute Engine](#)

- M7 Resource Monitoring



=

[Essential Google Cloud Infrastructure: Core Services](#)

- M4 Resource Monitoring



Skill Badges



Google Cloud

[Set Up an App Dev Environment on Google Cloud](#)



Google Cloud

[Develop your Google Cloud Network](#)



Documentation

[Managing metric-based alerting policies | Cloud Monitoring](#)

[Introduction to alerting | Cloud Monitoring](#)

Knowledge Check 1

What GKE object implements an http(s) load balancer?

- A. Service
- B. Pod
- C. Deployment
- D. Ingress



Knowledge Check 1

What GKE object implements an http(s) load balancer?

- A. Service
- B. Pod
- C. Deployment
- D. Ingress



Knowledge Check 2

Which Cloud Run autoscaling setting should you set if you want to limit cost?

- A. Min Instances
- B. Max instances
- C. Concurrency settings
- D. CPU utilization



Knowledge Check 2

Which Cloud Run autoscaling setting should you set if you want to limit cost?

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C. Concurrency settings

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